Scalability & Future Plan

Date	25 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Only supports local deployment	Cannot be accessed remotely by farmers	High
2	No user authentication or history	Users cannot save or track past predictions	Medium
3	Limited to one dataset (Kaggle)	Model may not generalize to all regions/ climates	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single user on local Flask server	Deploy on cloud (Render/AWS) with Gunicorn WSGI server
2	Data Storage	CSV file + Pickle model	PostgreSQL database for user data, predictions and history
3	Performance	Local inference only	API caching, model optimization, batch predictions
4	Security	No authentication	Add user registration, JWT tokens and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Cloud deployment (Render/Heroku) + User auth	2 months	Remote access for farmers, personalized experience
Phase 3	Regional dataset integration + Multi-language UI	4 months	Better predictions for diverse regions, wider user base
Phase 4	Mobile app (React Native) + Weather API integration	6 months	Real-time weather-based recommendations, mobile accessibility